

PROJECT AND TEAM INFORMATION

Project Title

UNIGO-A Smart Campus Navigation and Student Assistance Platform

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

* There are numerous challenges that occur on a regular basis, which appear to be minor but difficult to handle by college going people. For an instance ,it is seen that the students seems to have difficulty locating their classes ,labs and so many other places within the campus. Despite our time in within the campus, there are always places we haven't discovered or know about. And another concern is to find the bus base of the route you want to take as there are currently 22 open bus routes covering major locations and it is pretty difficult to know which base your bus depart.

* Another challenge appears to be finding the lost object or its owner. Usually students trying to find their stuff just wander aimlessly hoping to find it and same goes for the person who finds it. They all try to contact friends and try messaging on groups,hoping too find some clues ,which means the process of receiving help is slow and ineffective.

* Student and visitors faces daily frustrations with broken or poor campus facilities but we have no way to voice our complaints.

* All these challenges, may appear small, but consumes a lot of time and leading to an insufficient dependency on someone else.

* Therefore our idea is to create an online platform which would be very helpful in tackling such kind of problems.

Consequently to resolve these challenges ,we've decided to make a campus guide website which will help the college community to make their college experience a breeze.

State of the Art / Current solution

- We've decided to make this platform after going through the hassle of painfully looking for bus stops and classrooms, losing items and much more.
- During the start of the year, a new student entering the campus doesn't know where to look for solutions to these issues. Asking seniors, professors or guards rarely proves fruitful as everyone is busy or they themselves don't have an exact guide.
- Bus stops change frequently, and the only solution is to wander and ask for directions. Losing a valuable item is the worst challenge of all, and though lost and found exists, mostly it is not as quick or efficient, and items are only retrieved at the end of day's cleaning.
- Our website will make all of these processes less toilsome and save time and effort for students, especially new ones.

Project Goals and Milestones

- The main objective of the project is to create a centralized platform for helping the students in dealing with routine problems faced on a college campus.

Main objectives are as follows-

- 1 To create an interactive college map which will contain all the main locations of the campus like classrooms, laboratories, washrooms, seminar halls, offices, buildings, facilities, and bus stops.
- 2 Enabling a student to locate and navigate to campus facilities using an interactive 2D map.
- 3 Providing the bus information through pre loaded bus schedules, guiding students to their respected staging area.
- 4 Creating a Lost and Found system where students will be able to register their lost item and report any finds through descriptions, images, category and campus location.
- 5 Catering the students to give useful reviews and tips regarding various location on the campus.
- 6 Building up the system so that the other facilities and services available on the campus can be integrated in the map in the future.

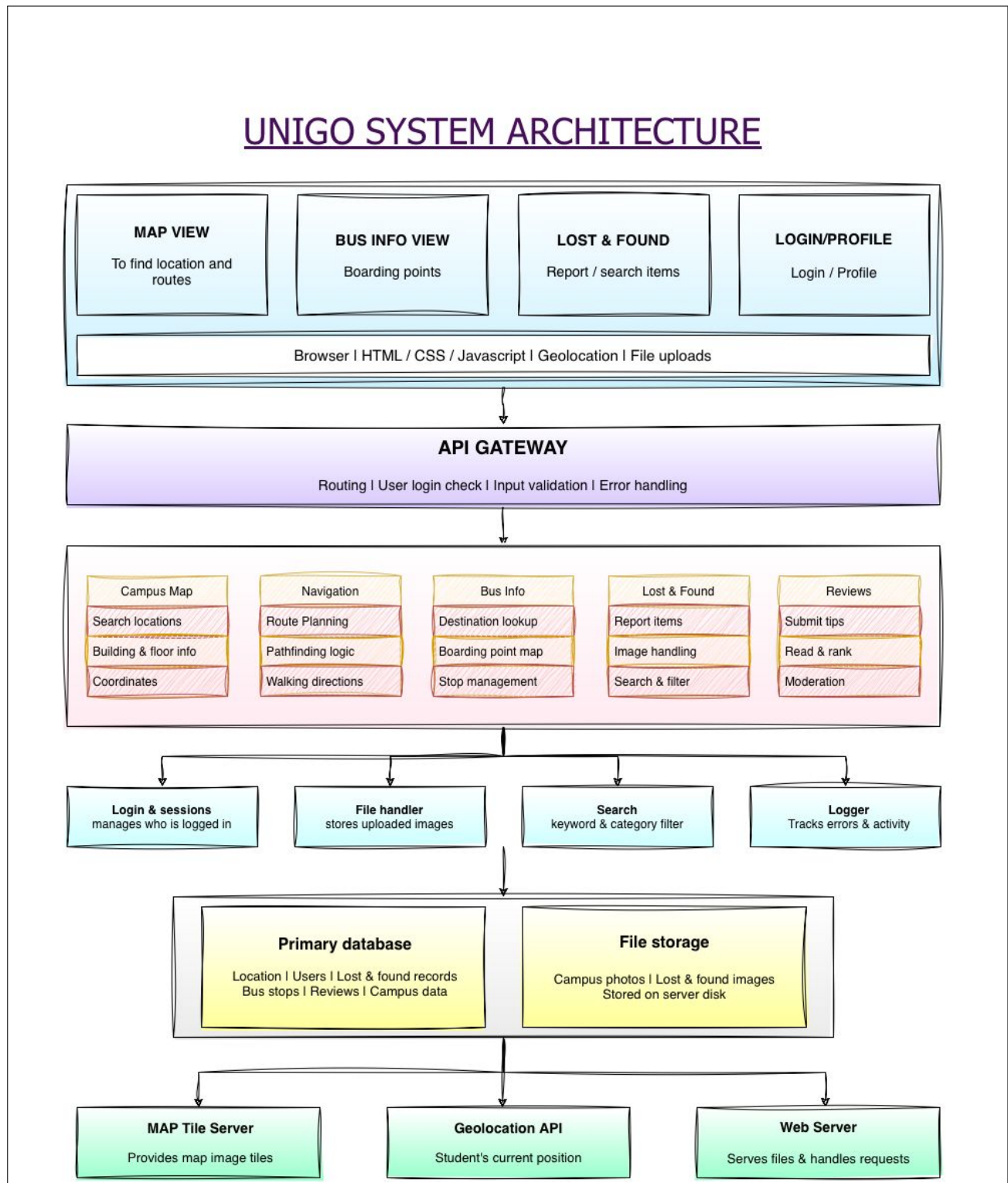
-Early milestones:

- . Gather all necessary information related to location, facilities and campus bus.
 - . To plan the design of website's interface and database.
 - . To create an interactive campus map and location search system.
 - . Implement navigation and current location support.
 - . Create Lost and Found system.
 - . Adding campus bus information.
 - . Add student reviews and location information.
 - . We will Test the system with students and improve usability through their feedback.
- Future milestones may include the features of 3D campus visualization, smart matching of Lost and Found system, accessible navigation and many more.

Project Approach

- Our project 'UNIGO' will be developed as a web - based platform, providing an accessible user interface for students to access various facilities such as navigating the campus, searching for classrooms, a lost and found feature and reviews for the betterment of the college.
- The website will be built using HTML, CSS and Javascript, while we'll operate using C++ in the backend. Information like the locations of classrooms and buses, lost and found image submissions and more will be stored in a database of our website's server.
- The campus map serves as our central feature, showcasing various campus locations connected via nodes, providing exact pathways to navigate between them. Information regarding buses will be manually curated, mapping all boarding points. The Lost and Found feature lets a student upload images, descriptions and categories of objects along with the campus location.
- UNIGO will be designed in a modular manner, so more features can be further incorporated without changing the base program. Each feature will operate independently, so that we can easily distribute work among the team members; the core program as of now focuses on basic functionality such as navigation, search and review. For future stages, smarter features like personalized notifications, timetable change alarm, crowd heat-mapping and more can be added.

System Architecture



Project Outcome / Deliverables

-The main outcomes /deliverables for this project include the following:

- 1.Campus Map and Navigation: An interactive map will show the important campus places like classrooms, laboratories, offices, restrooms, professor's cabins, first-aid area and bus stops. Along with this a navigation and search option will be added for students and others through which they can search for their required place and get a suitable route from their current location.
- 2.Bus Route Information: The system will display the information about buses, their different boarding points , tells the destinations of selected bus, with their numbers and departure timings based on the collected data.
- 3.Lost and Found Management: To identify the items , students can add relevant details of their lost or found things like images, description and location to detect the item related area.
- 4.Student Feedback and Reviews:Allows students to share their feedback, reviews and observations about all the campus locations and facilities, including necessary changes that needs to be done.
- 5.Campus Database and Web Interface: For future updation, a protected database will store campus locations and facility information and will have a user-friendly interface that can be accessed through any devices without causing technical errors.
- 6.Overall Campus Assistance: The project will bring these features together on one platform to help students, teachers, and first-time visitors to locate required places, to find the bus information and manage Lost and Found information.

Assumptions

- We assume that students and visitors will have a phone, internet connection, and knows their exact location to use the Campus Guide. Bus route and timing information will also be available and updated when needed. Students will provide the correct details while searching for lost or found items and as well while filling the student information section. Users will use the platform properly and give genuine reviews and feedback based on their experience.

References

- Material Provided by our faculty, all the lecture notes, referenced books.

*GeeksforGeeks

*google maps